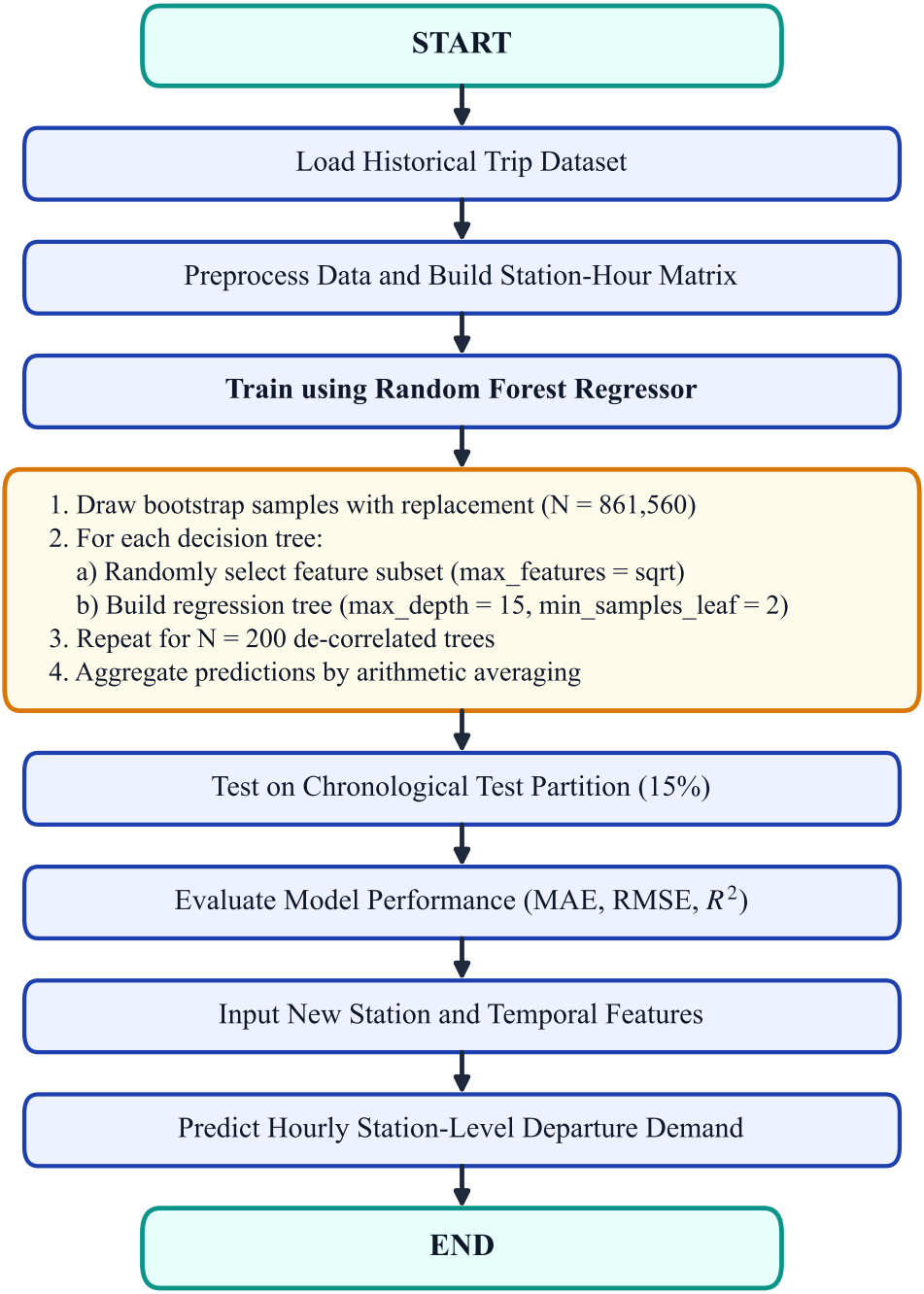
START
Load Historical Trip Dataset
Preprocess Data and Build Station-Hour Matrix
Train using Random Forest Regressor
1. Draw bootstrap samples with replacement (N = 861,560)
2. For each decision tree:
a) Randomly select feature subset (max_features = sqrt)
b) Build regression tree (max_depth = 15, min_samples_leaf = 2)
3. Repeat for N = 200 de-correlated trees
4. Aggregate predictions by arithmetic averaging
Test on Chronological Test Partition (15%)
Evaluate Model Performance (MAE, RMSE, R^2)
Input New Station and Temporal Features
Predict Hourly Station-Level Departure Demand
END